C# OOP Foundations & ADTs Examination

Directions: Read each question carefully. For the Multiple Choice section, select the best answer. For the Identification section, write the correct term or keyword that matches the description.

Part 1: Multiple Choice (1–25)

1. Which of the following best describes a class?
   A. A concrete entity created in system memory (Heap).
   B. A template or definition that describes what its instances will contain and do.
   C. A data type that always stores data values directly on the Stack.
   D. A specific method used to clear runtime memory.

2. Where in system memory is an object (the instance) created?
   A. Stack
   B. Cache
   C. Heap
   D. Hard Drive

3. In the architectural blueprint analogy provided in the text, what does the physical building represent?
   A. The Method
   B. The Class
   C. The Property
   D. The Object

4. What do properties represent in an object?
   A. Dynamic actions or verbs
   B. Memory address pointers only
   C. Characteristics, state, or nouns
   D. Internal interface contracts

5. Which of the following is an example of a method?
   A. AccountBalance
   B. NodeValue
   C. DepositMoney()
   D. IsFull

6. How do value types handle memory allocation when they are copied?
   A. They duplicate only the address pointer.
   B. They allocate completely new, independent memory spaces.
   C. They point directly to the Heap memory.
   D. They share the same memory space to save runtime data.

7. Which of the following is classified as a reference type?
   A. int
   B. struct
   C. double
   D. interface

8. Pressing a button to brake a vehicle without knowing hydraulics mechanics is an example of:
   A. Data Abstraction
   B. Control Abstraction
   C. Encapsulation
   D. Polymorphism

9. Accessing database records without managing physical arrays or files is an example of:
   A. Data Abstraction
   B. Control Abstraction
   C. Inheritance
   D. Value Typing

10. What is an Abstract Data Type (ADT)?
   A. A concrete array implementation with tracking variables.
   B. A theoretical design model specifying what data operations are available, completely ignoring how they are coded.
   C. A C# feature that automatically forces fields to be public.
   D. A method that runs on the Stack to copy value types.

11. Which C# construct is used as the primary vehicle to represent a conceptual ADT model?
   A. struct
   B. class
   C. interface
   D. virtual

12. According to the text, what behavioral rule governs a Stack ADT?
   A. First-In, First-Out (FIFO)
   B. Last-In, First-Out (LIFO)
   C. Random Access Allocation
   D. Secure State Shielding

13. In the provided ArrayStack implementation, what is the default size allocated for _storage?
   A. 10
   B. 50
   C. 100
   D. 0

14. What are the core architectural design guidelines for resilient software engineering called?

A. The Four Pillars of OOP
B. The Control Abstractions
C. The Reference Types
D. The ADT Frameworks

15. Hiding data inside an object and exposing it only via methods and properties defines which pillar?
A. Abstraction
B. Encapsulation
C. Inheritance
D. Polymorphism

16. Allowing custom derived classes to treat or morph shared behaviors differently defines which pillar?
A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism

17. Reusing properties and methods from existing parent classes defines which pillar?
A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism

18. In C# encapsulation, how should internal state fields be restricted?
A. They should be set to public.
B. They should be set to private.
C. They should be set to abstract.
D. They should be set to virtual.

19. What serves as the "gateway" to safely expose and protect fields from illegal outside changes?
A. C# Properties
B. Stack pointers
C. Abstract classes
D. Value Types

20. Which pillar reduces the cognitive load on developers by hiding messy, background engine design logic?
A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism

21. What type of relationship hierarchy does Inheritance establish?
    A. "Has-A" relationship
    B. "Uses-A" relationship
    C. "Is-A" relationship
    D. "Implements-A" relationship

22. If class Admin inherits from class User, which class is the parent base class?
    A. Admin
    B. User
    C. IStack
    D. DatabaseConnector

23. What is the literal meaning of Polymorphism?
    A. Code reuse without duplication
    B. The ability of an object to take on many forms
    C. Data shielding from outside manipulation
    D. Allocation of memory on the Stack

24. Which pair of keywords is used in C# to achieve dynamic runtime change via method overriding?
    A. public and private
    B. abstract and interface
    C. virtual and override
    D. class and struct

25. If Shape s = new Circle(); is executed, and both classes define a Draw() method correctly using polymorphism, what will s.Draw(); print?
    A. Shape
    B. Circle
    C. Base
    D. Nothing, it will cause a compiler error.

Part 2: Identification (26–40)

26. The specific system memory region where value types store their data values directly.

27. The functions defined inside a class that perform actions, execute operations, or alter an object's properties.

28. The template or blueprint that occupies absolutely no memory until it is instantiated.

29. The specific C# data type category that stores a memory address pointer pointing to heap memory.

30. The expected method of a Stack ADT used to add an item to the collection.

31. The expected method of a Stack ADT used to remove an item from the collection.

32. Hiding how underlying data records are structured and managed internally inside memory storage devices.

33. The C# keyword used in a parent class method declaration to allow it to be dynamically changed at runtime by a child class.

34. The C# keyword used in a child class method to replace or change the behavior inherited from a virtual parent method.

35. The relationship rule or phrasing (e.g., "A Dog ___ Animal") associated with inheritance.

36. The core pillar of OOP that bundles fields and methods together while restricting raw direct data access.

37. The modifier keyword applied to fields to keep them hidden and protected from direct external modification.

38. A concrete entity created in system memory that possesses its own unique state.

39. C# keyword used to define a blueprint that cannot be instantiated directly and forces subclasses to handle implementation details.

40. The primary structural benefit of inheritance that eliminates the need to duplicate code assets.